

Module 04: Timestamp Ordering Protocols

AKTU BCS501 | Unit 5: Concurrency Control Techniques

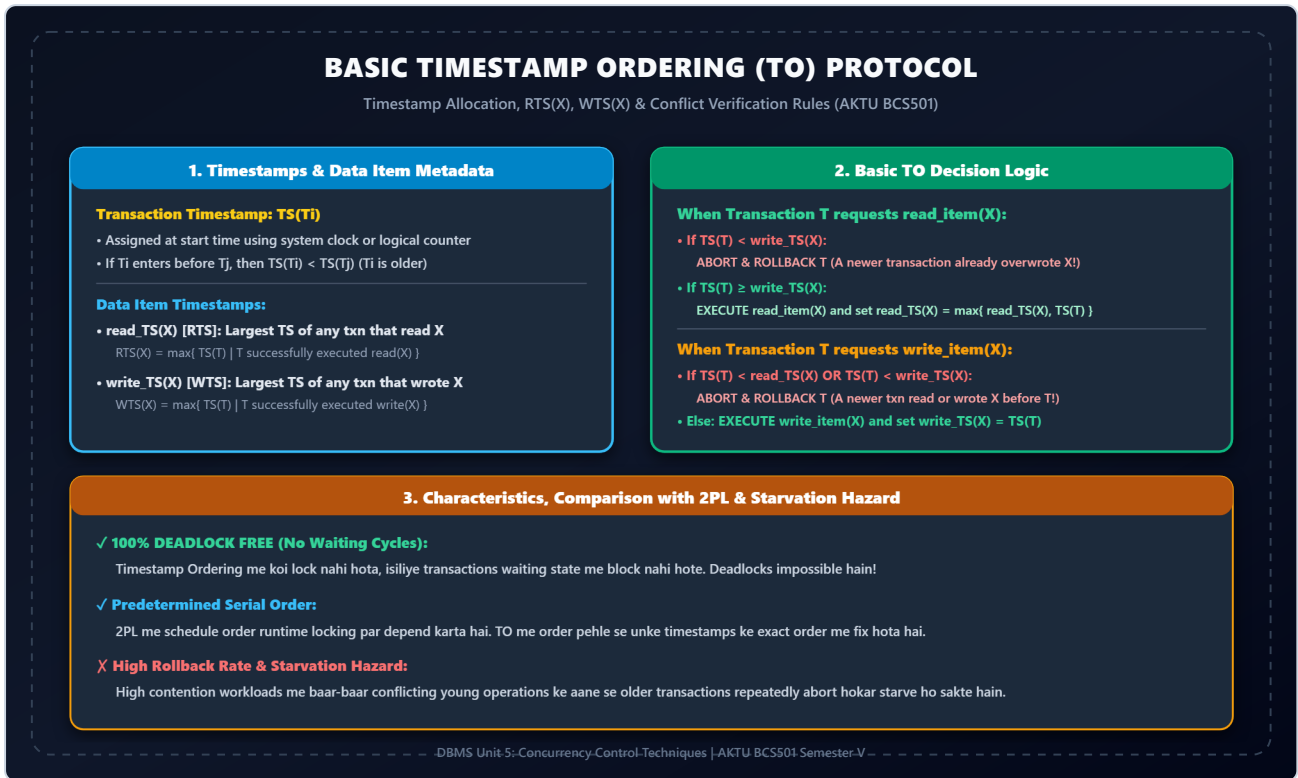


Figure 4.1: Basic Timestamp Ordering Rules & Metadata Architecture

1. Timestamp Assignment & Data Item Metadata

Every transaction receives a unique timestamp $TS(T_i)$ at start. Data items maintain $read_TS(X)$ (largest timestamp of any reader) and $write_TS(X)$ (largest timestamp of any writer).

2. Basic TO Rules Summary

Read and Write Check Conditions

- **read(X):** If $TS(T) < write_TS(X) \rightarrow$ ABORT T. Else execute and set $read_TS(X) = \max(read_TS(X), TS(T))$.
- **write(X):** If $TS(T) < read_TS(X)$ or $TS(T) < write_TS(X) \rightarrow$ ABORT T. Else execute and set $write_TS(X) = TS(T)$.

3. Deadlock-Free Guarantee vs Cascading Aborts

Because conflicting transactions are immediately aborted rather than forced to wait in queues, Timestamp Ordering is 100% deadlock-free. However, restart overhead can be high under contention.